

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Turbidity	WQP 006	1.0	

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1. Title

Standard Operating Procedure for Seawater Turbidity Nephelometric Determination

2. Applicability

This SOP applies to the determination of seawater Turbidity at shallow coral reef sites around Puerto Rico.

3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze coral reef seawater surface and bottom seawater Turbidity at shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater Turbidity data following the Quality Assurance Project Plan.

4. Method

- 4.1. EPA Method 180.1 (Environmental Protection Agency)

5. Health And Safety

- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2. Read each reagent SDS prior to use for safe handling and disposal.
- 5.3. Discard Formazin standard into the corresponding hazardous waste container.
- 5.4. Wear disposable latex gloves and safety glasses while performing this procedure.
- 5.5. Avoid inhalation, ingestion, and skin contact with Hydrazine sulfate (Formazin).

6. General Precautions

- 6.1. Always maintain two points of contact when deploying the Van Dorn bottle on the boat to avoid losing the bottle overboard.
- 6.2. Ensure the boat engines are idle and the captain approves of the Van Dorn bottle being deployed overboard before deploying the bottle to ensure that the rope does not become entangled with the boat propeller.
- 6.3. Do not use solvents, alkaline or abrasive cleaners, or other abrasives such as steel wool or scouring powder on the turbidimeter.
- 6.4. Always use clean and undamaged cells on the turbidimeter.
 - 6.4.1. Discard any scratched or damaged cells.
- 6.5. Let samples reach ambient temperature before filling the turbidimeter cell to avoid clouding of the cell.
- 6.6. Do not dilute the sample.
- 6.7. Do not shake calibration standards as this can introduce errors to the reading.
- 6.8. Always close the compartment lid during reading, calibration verification, and storage.
- 6.9. Remove the cell and batteries if the equipment will not be used and stored for a long period

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(more than one month) to avoid damaging the battery contacts.

6.10. Avoid water near the turbidimeter's battery compartment.

6.10.1. In case water gets into the compartment, turn off the equipment and remove the batteries. Dry the inside of the compartment and reinsert the batteries.

7. Training

7.1. Personnel shall be trained in the proper sample collection, handling, and analysis procedure for the measurement of seawater Turbidity.

7.2. Personnel shall be trained in the proper use of the Orion AQ4500 Portable Turbidimeter.

7.3. Personnel shall receive training on how to complete the field sampling log for this SOP.

8. Required Materials

- 8.1. Van Dorn Bottle
- 8.2. Orion AQ4500 Portable Turbidimeter
- 8.3. Distilled water
- 8.4. Soft, lint-free cloth
- 8.5. Disposable latex gloves
- 8.6. Batteries (4 AA Alkaline)
- 8.7. 1 Polypropylene Plastic Bottle for sample (250 mL)
- 8.8. 1 glass sample cell (10 mL) with cap
- 8.9. Calibration standard cells (5)
- 8.10. Calibration standards (AQ4500 standard solutions)
- 8.11. Field Form (Appendix F)

9. Procedure

9.1. Sample Collection

- 9.1.1. Prepare a Van Dorn bottle to be deployed.
- 9.1.2. Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3. Drop the bottle messenger to close Van Dorn at the desired depth.
- 9.1.4. Pull the Van Dorn bottle back up into the boat.
- 9.1.5. Connect the tubing to the Van Dorn bottle spigot to transfer water.
- 9.1.6. Push open the Van Dorn bottle's spigot valve to allow water flow.
- 9.1.7. Rinse three times a 250 mL plastic bottle for sample collection with water from the Van Dorn bottle.
- 9.1.8. Collect the sample until the bottle overflows.
- 9.1.9. Place the sample bottle inside the cooler until analysis in the lab.
- 9.1.10. Process samples within 48 hours.

9.2. Preparation of Secondary Calibration Standard Vials (AQ4500 calibration standards)

- 9.2.1. Ensure calibration standards have not reached expiration date.
 - 9.2.1.1. Shelf life is 12 months.

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- 9.2.2. Inspect vials to be used for calibration.
 - 9.2.2.1. Vials should not be scratched or dirty.
 - 9.2.2.2. Always manage vials from the top.
 - 9.2.2.3. One separate vial is to be used for each standard solution.
- 9.2.3. Rinse the inside of the vial twice with the calibration standard.
 - 9.2.3.1. Gently pour about 5mL of the standard into the vial.
 - 9.2.3.2. Cap and gently swirl the vial.
 - 9.2.3.3. Discard the rinse in the designated hazardous waste container.
- 9.2.4. Gently fill the vial with the standard up to the mark and cap the vial.
- 9.2.5. Repeat steps 9.2.3 and 9.2.4 for the remaining standard solutions.
 - 9.2.5.1. CAL 1: 0 NTU
 - 9.2.5.2. CAL 2: 1 NTU
 - 9.2.5.3. CAL 3: 10 NTU
 - 9.2.5.4. CAL 4: 100 NTU
 - 9.2.5.5. CAL 5: 1000 NTU
- 9.2.6. Perform one of the following as corresponding:
 - 9.2.6.1. Initial Calibration as described in step 9.3
 - 9.2.6.2. Daily Calibration Check as described in step 9.4
 - 9.2.6.3. Store filled calibration vials in the instrument case for future daily calibration checks.

9.3. Initial Calibration for EPA 180.1 Mode

- 9.3.1. Turn on the the Orion AQ4500 Turbidimeter
- 9.3.2. Select measurement mode: EPA 180.1.
- 9.3.3. Press #8 key (“cal”).
 - 9.3.3.1. “H2O INSERT” will be displayed.
- 9.3.4. Insert vial containing pure water and press “yes” key.
 - 9.3.4.1. Ensure the white triangle on the vial is aligned with the tab on the turbidimeter and the sample is fully inserted.
- 9.3.5. Wait for the following display: “H2O WAIT” and then “1.00 YES?”.
- 9.3.6. If the standard is 1.00, insert the standard vial and press the “yes” key.
 - 9.3.6.1. If the standard is another value, press “6” or “3” keys (up & down arrows), “CHANGE?” will be displayed.
 - 9.3.6.2. Press “yes” key.
- 9.3.7. Wait for the following display “STD VAL?”
- 9.3.8. Enter value of standard using the numeric keypad.
- 9.3.9. Press “yes” key to accept.
- 9.3.10. Repeat steps 9.3.6 to 9.3.9 for the remaining standards.
 - 9.3.10.1. When calibration is completed, the instrument will say “Cal Done” proceed to the measurement mode.
- 9.3.11. Perform a meter performance check.
 - 9.3.11.1. Refer to Daily Calibration Check instructions described in Section 9.4.

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9.4. Daily Calibration Check for EPA 180.1 Mode

Measure the calibration secondary standards and a blank (turbidity free water).

- 9.4.1. Place the turbidity meter on a flat and leveled surface.
- 9.4.2. Insert CAL 1 standard into the sample chamber.
 - 9.4.2.1. Press the vial down until it slides fully into the instrument.
- 9.4.3. Cover the vial.
 - 9.4.3.1. Ensure the white triangle on the vial is aligned with the tab on the turbidimeter and the sample vial is fully inserted.
- 9.4.4. Press #9 key (“meas”) to select measurement mode.
- 9.4.5. Record the reading in the Turbidity station lab notebook.
 - 9.4.5.1. Good performance: blank reads less than 0.1 NTU and standards read within 10% of the expected values.
 - 9.4.5.2. If results are not within the limits, repeat the check.
 - 9.4.5.3. If the results continue out of desired range, discontinue analysis and determine the cause of error.
- 9.4.6. Repeat steps 9.4.2 to 9.4.5 for the remaining standard solutions (CAL 2 to CAL 5).
- 9.4.7. Store filled calibration vials in the instrument case for future daily calibration checks.
 - 9.4.7.1. If the calibration standards no longer read within 10% of the nominal NTU value for each standard, the standard most likely has degraded.
 - 9.4.7.2. Refer to “Refilling the Calibration Standard Vials with Fresh Standard” Chapter 4, page 19 of the instrument user manual.

9.5. Sample Turbidity Measurements

- 9.5.1. Allow samples to reach room temperature.
- 9.5.2. Mix the sample thoroughly to disperse solids and wait until air bubbles disappear.
- 9.5.3. Pour the sample into a clean and dry turbidity vial.
- 9.5.4. Cap the vial.
- 9.5.5. Rinse the vial with a small amount of sample and empty.
- 9.5.6. Fill the sample to the white line.
- 9.5.7. Press #4 key “avg” to select a longer sample integration time of measurements.
- 9.5.8. Gently invert the vial two to three times to ensure the suspension is uniform.
- 9.5.9. Wipe the vial with a lint- free wipe to dry and/or remove fingerprints.
- 9.5.10. Place the vial into the sample chamber of the turbidimeter and cover it with the vial cover.
 - 9.5.10.1. Ensure the white triangle on the vial is aligned with the tab on the turbidimeter and the sample vial is fully inserted.
- 9.5.11. Press #9 key (“meas”) to measure.
- 9.5.12. Record the reading in the Turbidity station lab notebook or log the data.
 - 9.5.12.1. Report turbidity readings, to the nearest NTU, according to the below table from EPA 180.1. For example, a 5.07 NTU reading on the instrument would be reported as 5.05 NTU.

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Turbidity Range	Report to the Nearest
<u>NTU</u>	<u>NTU</u>
0 - 1.0	0.05
1 - 10	0.1
10-40	1
40 - 100	5
100 - 400	10
400 - 1000	50
>1000	100

9.5.13. Proceed with the next sample.

10. Documentation

- 10.1. Field data will be recorded in the Sampling Log (Appendix F).
- 10.2. Calibration data and sample turbidity readings will be recorded in the Turbidity station lab notebook.
- 10.3. Field and Calibration forms shall be kept in the designated file cabinet at CCRI office.
- 10.4. Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 10.5. All spreadsheets and depth profile logs should be uploaded to the internal CCRI Water Quality Database.
- 10.6. All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

- 11.1. Wash sample bottles with laboratory detergent and rinse thoroughly with distilled water and allow to air dry.
- 11.2. In case of scratches on vials, lightly cover the vials with silicone oil and spread the excess using a lint-free wipe.
 - 11.2.1. Discard vials if the silicone oil does not improve the surface.
- 11.3. Clean turbidimeter external surfaces with a dry or slightly damp cloth, free of lint and dust as necessary.
 - 11.3.1. Do not use solvents.
- 11.4. Remove the cell and batteries if the equipment will not be used and stored for a long period (more than one month).

12. Troubleshooting

- 12.1. If the calibration standards no longer read within 10% of the nominal NTU value for each standard, the standard most likely has degraded.
 - 12.1.1. Refer to “Refilling the Calibration Standard Vials with Fresh Standard” Chapter 4, (page 19) of the instrument’s user manual.
- 12.2. Refer to page 30 of the User Manual to troubleshoot error codes in the meter display.

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13. QA/QC

- 13.1. As per QAPP Table 4, seawater turbidity should be measured in Nephelometric Turbidity Units (NTU) with an accuracy of $\leq \pm 20\%$ and a precision of $\leq 20\%$.
- 13.2. Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.

14. References

- 14.1. U.S. ENVIRONMENTAL PROTECTION AGENCY. 1993. Methods for Determination of Inorganic Substances in Environmental Samples; EPA-600/R/93/100—Draft. Environmental Monitoring Systems Lab., Cincinnati, Ohio.